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(See 6.12)

MILITARY SPECIFICATION

COOLER, DRINKING WATER, MECHANICALLY REFRIGERATED, SELF-CONTAINED, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-contained, mechanically refrigerated, drinking water coolers for Naval shipboard use. Only one classification of cooler is covered by this specification.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-P-664 - Primer Coating, Alkyd, Corrosion-Inhibiting, Lead and Chromate Free, VOC-Compliant.

PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4110

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MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-S-901	- Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-E-2036	- Enclosures for Electric and Electronic Equipment.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-I-24172	- Insulation, Plastic, Cellular Polyurethane, Rigid, Preformed and Foam-in Place.
MIL-C-81302	- Cleaning Compound, Solvent, Trichlorotrifluoro-ethane.

STANDARDS

MILITARY

MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278	- Welding and Casting Standard.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.
DOD-STD-1399, Section 301	- Interface Standard for Shipboard Systems Section 301 Ship Motion and Attitude. (Metric)
MIL-STD-2073-1	- DOD Materiel Procedures for Development and Application of Packaging Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0900-LP-001-7000	- Fabrication Inspection of Brazed Piping Systems.
0908-LP-000-3010	- Surface Ship Shock Design Criteria.

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AIR-CONDITIONING AND REFRIGERATION INSTITUTE (ARI)

1010 - Standard for Drinking Fountains and Self-Contained Mechanically-Refrigerated Drinking-Water Coolers. (DoD adopted)

(Application for copies should be addressed to the Air-Conditioning and Refrigeration Institute, 1501 Wilson Blvd., Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B18.6.4 - Thread Forming and Thread Cutting Tapping Screws and Metallic Drive Screws, (Inch Series). (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip. (DoD adopted)

A 269 - Standard Specification for Steamless and Welded Austenitic Stainless Steel Tubing for General Service. (DoD adopted)

B 36 - Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar. (DoD adopted)

B 88 - Standard Specification for Seamless Copper Water Tube. (DoD adopted)

B 135 - Standard Specification for Seamless Brass Tube. (DoD adopted)

B 152 - Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar. (DoD adopted)

B 280 - Standard Specification for Seamless Copper Tube for Air Conditioning and Refrigeration Field Service. (DoD adopted)

C 177 - Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded-Hot-Plate Apparatus. (DoD adopted)

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W. Washington, DC 20037.)

UNDERWRITERS LABORATORIES, INC. (UL)
UL 399 - Drinking-Water Coolers.

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.2.

3.2 Materials. Materials, parts, and components shall conform to the requirements specified herein. Materials, parts, and components forming a part of the finished product shall be new and of adequate size or capacity for the intended purpose. Materials shall be free from any defects that might affect the serviceability or appearance of the finished product. Where materials are not specified, they shall be selected on the basis of successful state-of-the-art commercial applications for service conditions similar to those specified herein (see 6.3).

3.2.1 Hazardous materials. Materials which produce dangerous gases or other harmful effects under conditions (including fire) encountered in Naval shipboard service shall not be used without specific approval of the contracting activity. Magnesium and its alloys shall not be used. No instruments containing mercury shall be used in the manufacture or testing of any equipment destined for installation on a ship (see 6.3).

3.2.2 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new. Recovered materials may be used provided the intended use is not jeopardized. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

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3.2.3 Corrosion protection. Corrosion-resisting steel, copper, and brass referenced herein are considered satisfactory corrosion-resisting materials. Where corrosion-resisting steel is specified, it shall conform to class 302 (UNS S30200) or class 304 (UNS S30400) of ASTM A 167 and where exposed in final assembly shall have a number 4 or 6 finish. Parts other than corrosion-resisting materials shall be protected against corrosion after fabrication. Paints shall not be used where disassembly for maintenance or repair will reduce or eliminate the effectiveness of the corrosion protection.

3.2.3.1 Attaching hardware. Screws exposed in final assembly shall be corrosion-resisting steel. Bolts, nuts, pins, screws and other fastenings shall be of the same material being jointed or supported or shall be cathodic to the materials being joined or supported. Self-tapping spaced thread screws of the thread-forming type (see ANSI B18.6.4, types A, B and BP) will not be permitted.

3.3 Reliability. The mean-time-between-failure (MTBF) for each water cooler shall be not less than 10,000 operating hours (see 6.3).

3.3.1 Water cooler failure. Failure of a water cooler shall be defined as any event which requires it to be shut down for corrective maintenance requiring more than 2 hours to perform. Any corrective maintenance that may be completed by a ship's force in less than 2 hours and is supported by onboard repair parts and tools shall not be considered a failure.

3.3.2 Service life. The water cooler and its components shall have a trouble free operating life of not less than 150,000 hours when subjected to any or all of the shipboard conditions specified in 3.6.2. Such performance shall be obtained with a minimum and reasonable degree of preventive and corrective maintenance.

3.4 General. The drinking water cooler shall consist of a compressor, air-cooled condenser, water storage tank with evaporator coil, heat interchanger, dehydrator, bubbler valve and stream regulator. A water pre-cooler and refrigerant liquid receiver are optional for use. Additionally it shall comprise refrigerant piping and water and drain piping, together with temperature controls and miscellaneous accessories. An integral frame shall support the various assemblies and components. The drinking water cooler shall be self-contained, meet the requirements of ARI 1010 and shall perform as specified herein. Coolers shall operate on 115-volt, single-phase, 60-Hertz (Hz) alternating current (ac) and meet the requirements of UL 399.

3.5 Construction.

3.5.1 General assembly. The equipment shall operate under shipboard conditions of vibration, pitch, roll, and shock. The construction of the equipment shall conform to the requirements of UL 399. The drinking water cooler shall be not greater than 17 inches square and 44 inches in height from the top rim of the basin to the base. Wood shall not be used in the equipment construction. The drinking water cooler shall be factory assembled, complete,

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self-contained, with the refrigerating system dehydrated and charged with the proper operating quantities of refrigerant and oil. The unit shall be ready for operation after removal of shipping protection, opening of valves, connection to services and attachment of accessories (see 6.3).

3.5.2 Safety. To ensure maximum protection for personnel against hazards and accidents, the drinking water cooler shall incorporate the requirements of UL 399 and have the following safeguards and protective features:

(a) Components with exposed operating surface temperatures in excess of 125 degrees Fahrenheit (°F) that are within easy grasp, or that are susceptible to being inadvertently touched during maintenance tasks, shall have those high temperature surfaces suitably insulated or protected with metallic guards.
(b) Exposed edges shall be generously rounded and made smooth to prevent cutting edges and sharp corners.
(c) Insulated connectors and plugs of the electric current-carrying conductors shall be constructed in such a way that they will not constitute an electrical shock or residual discharge hazard when they are disconnected for maintenance and repair.

3.5.2.1 Lead free requirements. To ensure protection against health hazards, the drinking water cooler shall be constructed of "lead-free" pipe, solder, and flux. The term "lead free" is defined as: (a) when used with respect to solders and flux, lead content shall be not greater than 0.2 percent. (b) when used with respect to pipes and pipe fittings, lead content shall be not greater than 8 percent.

3.5.3 Cabinet. The integral frame shall be constructed of welded structural members. Four 1/2-inch diameter bolt holes shall be provided in the structural members forming the base for bolting the unit to a deck. The holes shall be located in each corner of the cooler spaced 12-1/4 inches center to center. The frame assembly shall be made from corrosion resistant steel or be galvanized.

3.5.3.1 Cabinet top. The cabinet top shall be fabricated from corrosion-resistant steel and drawn in one piece to form a basin. The sides of the top shall overlap the front, sides, and back of the cabinet for water shed. The cabinet top shall have a drain opening with a strainer in the basin and shall be provided with openings to permit the mounting and connecting of a bubbler valve and pitcher filler. The basin strainer openings shall be not greater in area than the diameter of 1/4-inch holes. The front and sides of the unit shall be fabricated from corrosion-resistant steel not less than 20 gauge (American Wire Gauge (AWG)) and may be formed either as a wrapper or from individual panels. Gas or electric arc welded joints, visible in final assembly, shall be ground or polished to match the surfaces adjacent to the welds.

3.5.3.2 Access. Valves, controls, and equipment subject to service and repair shall be readily accessible for servicing through a removable front panel. Where panels are provided for access to equipment, machine screws may be used for fastening.

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3.5.3.3 Ventilation. Ventilation shall be provided for condensing unit equipment. Where ventilation is provided in the front or sides of the cabinet, the opening shall be louvered type. Slotted openings shall be arranged for water shed. Louvered openings for ventilation shall not exceed 1/2-inch in clear width and shall not have sharp or ragged edges. Electrical devices shall be located so as to be protected or shall be shielded from water spillage or drippage.

3.5.4 Condensing unit. Unless otherwise specified (see 6.2), the condensing unit shall operate with dichlorodifluoromethane (R-12) refrigerant and shall consist of a compressor, condenser, receiver (if required), fan, motor and accessories mounted on a fabricated steel frame.

3.5.4.1 Compressor. The compressor shall be of the hermetic type. The hermetic type compressor shall consist of a compressor and a motor enclosed in a welded shell. Provision shall be made for adequate lubrication of all rubbing and wearing surfaces, including operation under shipboard conditions of inclination and rolling. The capacity rating of the compressor, based on operation in an ambient temperature of 90°F, shall be not less than 3600 British thermal units (Btu) per hour under the conditions of 40°F evaporating temperature.

3.5.4.2 Condenser. The condenser shall be forced air cooled and constructed of finned tubes conforming to either ASTM B 280 or ASTM B 88. The condenser shall be not greater than five rows deep. Fins shall be soldered, welded, or mechanically bonded to the tubes and shall be not less than 0.006 inch thick and spaced not greater than 12 fins per inch. The condenser shall be treated for corrosion resistance after fabrication. The fan shall be located as nearly as is practicable to the center of the condenser face and, where necessary for uniform air flow, a fan shroud shall be provided. The condenser shall have sufficient cooling surface area to permit water cooler operation in an ambient temperature of 120°F with a maximum discharge pressure not greater than 220 pounds per square inch (lb/in^2) gauge, and an evaporating temperature of 40°F, when operated in accordance with 3.6.5.

3.5.4.3 Refrigerant liquid receiver. The refrigerant liquid receiver shall have an internal volume not less than 25 percent greater than the liquid volume of the complete refrigerant charge. The receiver shall contain a fusible plug for protection against excessive temperatures. The plug shall melt at a temperature not less than 250°F and not greater than 300°F.

3.5.4.4 Refrigerant piping. The refrigerant piping shall be fabricated from one or more of the following materials:

(a) Copper, ASTM B 280 (UNS C12200).
(b) Stainless steel, ASTM A 269 (UNS S30400), seamless.

Connections and joints shall be kept to a minimum. When copper piping is used, all joints or connections shall be silver brazed using fittings of wrought copper or forged brass. Stainless steel piping joints or connections shall be welded. Piping adjacent to frame members or to other components or parts shall be rigidly supported to prevent movement and rubbing as a result of environmental vibration. The suction line to the compressor within the machinery compartment shall be insulated to prevent condensation. Piping systems shall withstand the pressure and leak tests specified (see 4.3.1).

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3.5.4.4.1 Service valves. Self-sealing capped refrigerant service valves shall be installed in the compressor suction and discharge refrigerant piping.

3.5.4.5 Dehydrator. A dehydrator shall be provided in the liquid refrigerant circuit. Each dehydrator shall be equipped with an auxiliary screen or other protective means at the dehydrator outlet to prevent passage of dehydrating agent in the event of rupture of the cartridge screen outlet. The desiccant shall be silica gel, activated alumina, or of a type approved by the Command or agency concerned.

3.5.4.6 Heat-interchanger. A heat-interchanger shall be provided. Bonding of the liquid refrigerant line to the suction line to provide adequate liquid sub-cooling will be considered satisfactory as a heat-interchanger.

3.5.5 Cooling unit. The cooling unit shall consist of a water storage tank with a refrigerant coil and insulation all housed within the water cooler cabinet. The water storage tank shall be located within the cabinet so that there is not less than 2-1/2 inches of insulation between the tank and the sides, top, or bottom of the cabinet. The water storage tank shall be rigidly supported within the cabinet. Water supply and drain piping shall be arranged so as not to come in contact with the refrigerant piping or rub against the tank or cabinet. There shall be no piping joints within the cabinet.

3.5.5.1 Water storage tank. The water storage tank shall be either corrosion-resistant steel in accordance with ASTM A 167, brass in accordance with ASTM B 36, or copper in accordance with ASTM B 152. All tank seams and related tubing connections shall be welded, silver-brazed, or equivalent. Internal surface coatings are prohibited. Water outlet connections shall be arranged so that sediment may be readily flushed out. Means shall be provided to avoid short circuiting or warm incoming water to the outlet. One or more wells shall be provided in the top of the tank for insertion of the operating and low temperature safety thermostatic switch thermal bulbs. Where a single well is used, the operating bulb shall be located in the bottom of the well and the safety thermostat near the top or in the area where the lowest water temperature will occur. The water storage tank and water piping shall withstand the pressure and leak test specified in 4.3.3 and 4.3.4.

3.5.5.2 Refrigerant coil. The refrigerant coil shall be either corrosion-resistant steel in accordance with ASTM A 269 or copper in accordance with ASTM B 280. The corrosion-resistant steel tubing shall be internal (immersed in the water) and shall be welded to the tank wall at the points of entrance and exit with no internal joints. Where refrigerant coil is corrosion-resistant steel, the water storage tank shall also be corrosion-resistant steel. The coil for nonferrous tanks shall be copper, continuously bonded (minus 10 percent) to the external surface of the tank. The use of additional refrigerant for water cooling by means of tube-in-tube or tube-on-tube exchanger will not be permitted. Sufficient coil surface shall be provided to permit operation as specified in 3.6.3.

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3.5.6 Water and drain piping. Brass tubing shall be in accordance with ASTM B 135 and copper tubing in accordance with ASTM B 88. Internal water piping connections shall be silver brazed joints, using silver brazing alloys, or phosphorous copper. Fittings for external connections shall be secured to the frame or to a bracket secured to the frame in such a manner that, when connections are made to external water and drain lines, no strain will be placed upon the internal piping. Offsets shall be incorporated within the cooler on all lines to relieve stress due to vibration. Piping adjacent to frame members or other component parts shall be rigidly supported to prevent movement and rubbing as a result of environmental vibration. Fresh water supply, waste, and tank drain connections shall be provided. Water supply and tank drain connections shall be female 3/8-inch National taper pipe (thread) (NPT) or union with solder fittings provided with plugs. The waste connection shall be 1-1/4 inch female NPT or union with solder fittings provided with plug. The pitcher filler connections shall be male 3/8-inch NPT and provided with a brass, chrome plated, hexagon pipe cap. Water piping within the cabinet shall be insulated.

3.5.7 Bubbler valves. The bubbler and pitcher filler valves shall be self-closing and parts in contact with water, except gasket and seals, shall be made of brass. Exposed parts shall be chrome plated. The valves shall be constructed for rough usage.

3.5.7.1 Bubbler guard. The bubbler shall be sanitary and shall have a slanting jet, protected by a guard and configured to prevent contact with the mouth. The guard shall prevent foreign matter from dropping vertically into the orifice of the bubbler. The orifice of the bubbler shall be located not less than 3/4 inch above the overflow level of the top. The bubbler and guard shall be made of brass, chrome plated.

3.5.7.2 Pitcher filler. The pitcher filler, when required (see 6.2), shall be of brass, chrome plated and shall fill a pitcher 11 inches high. Waste water from the pitcher filler shall drain to the basin. The pitcher filler shall incorporate a self-closing valve and have a female 3/8-inch NPT connection for connecting to the drinking water cooler. The pitcher filler shall be packed separately for installation by the installing activity.

3.5.7.3 Stream regulator. An automatic stream regulator of brass material shall be provided in the bubbler line or shall be integral with the bubbler and shall operate on fresh water system pressure ranging from 20 to 70 lb/in^2. The regulator shall maintain a bubbler-stream height of not less than 1 inch nor greater than 3 inches above the top of the bubbler guard for any one setting without readjustment, and shall be readily accessible for adjustment or repairs.

3.5.8 Pre-cooler. A pre-cooler, where provided, shall be constructed of nonferrous seamless tubing. Both inlet and drain water passages shall be fully enclosed with two walls of metal separating the incoming drinking water from the drain water. There shall be no joints in the area of contact of the drain and supply water tubing. Surfaces in contact with the drain water shall be smooth and there shall be no internal restrictions or passages that would reduce the free opening to less than 5/8-inch diameter. The pre-cooler shall be securely mounted within the water cooler cabinet and shall be located not less than 1 inch from the water storage tank.

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3.5.9 Temperature control. The automatic control of the compressor operation shall be accomplished by means of a temperature control switch. The switch shall start and stop the compressor motor when the temperature of the water in the storage tank rises above or falls below predetermined values. The temperature control switch shall be set to maintain design water temperature. The switch shall include a stop to prevent the compressor from starting at a temperature lower than 35°F. Operation of the switch contacts under the shock test is permissible.

3.5.9.1 Control device access. Automatic refrigeration control devices shall be readily accessible for adjustment and servicing. Where thermostatic expansion valves are used, they shall be of the pressure limiting type. Where the compressor is a low starting torque type, the expansion valve shall have a bleed port for system pressure equalization.

3.5.10 Compressor motors. Alternating current motors shall be in accordance with NEMA MG 1 Definite-Purpose Machines (motors for hermetic refrigeration compressors). The compressor motor windings shall be protected against overload by a thermal protector built into the motor. The temperature rise of the hermetically sealed motor shall be not greater than 70 degrees Celsius (°C).

3.5.11 Fan motor. When a motor is used to drive a condenser cooling fan, the fan motor enclosure shall be dripproof. The motor shall have a sealed or enclosed lubricant system and shall not require bearing lubricant to be changed or supplemented periodically.

3.5.12 Motor starting relays. Motor starting relays, used in lieu of centrifugal switches for disconnecting the starting winding of single-phase, ac motors, shall be mechanically shockproof (as mounted on the refrigeration equipment when tested in accordance with MIL-S-901) and dripproof in accordance with MIL-E-2036. Rubber or other shock-mounts which are subject to deterioration in service shall not be used for mounting the relays.

3.5.13 Wiring. The wiring between the electric devices (motors, motor controllers, control devices) shall be in accordance with UL 399. The wiring shall be complete and the equipment shall be ready for connection of the incoming power cable at installation. A terminal block in accordance with MIL-E-917 for three-conductor power cable shall be provided for the incoming power cable.

3.5.14 Protection against electrolytic or galvanic corrosion. Water coolers shall be constructed with careful attention to protection against electrolytic or galvanic corrosion. Joints between dissimilar metals which may become wetted in service shall be protected against electrolytic or galvanic corrosion by proper selections of materials, plating, isolation, insulation, paint, area relationship, and gaskets in accordance with MIL-STD-889.

3.5.15 Thermal insulation. Thermal insulation materials, other than pipe insulation, shall have a thermal conductivity not greater than 0.30 Btu per hour per square foot per inch thickness per degree Fahrenheit temperature difference at a mean temperature of 50°F when tested by the guarded-hot-plate method in accordance with ASTM C 177. Insulation other than foam-in-place shall be applied

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in a manner to prevent settling in service, and provided with a vapor barrier to prevent the entrance and accumulation of moisture in the water cooling unit. Where molded segments are used, an adhesive shall be used to bond the segments together. Insulation spaces shall be filled so as to leave no voids. Foam-in-place insulation shall be in accordance with type II, class 1 of MIL-I-24172. Piping or surfaces that may form condensation shall be insulated to prevent dripping of moisture. Polystyrene insulation shall not be used.

3.5.16 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.5.17 Welding. Welding and allied processes, including inspection, repairs, and materials shall be in accordance with MIL-STD-278.

3.5.18 Brazing. Brazing and inspection shall be in accordance with NAVSEA 0900-LP-001-7000.

3.6 Performance.

3.6.1 Refrigerant pressure and leakage. The drinking water cooler shall withstand the refrigerant pressure and leakage test (see 4.3.1), refrigerant vacuum test (see 4.3.2), gas leakage test (see 4.3.3), and hydrostatic test (see 4.3.4) with no evidence of leakage.

3.6.2 Environmental performance. The drinking water cooler shall operate and perform its function without deleterious effects on reliability and wear rate when installed and operated on board Navy ships under shipboard environmental conditions, which may prevail simultaneously and occur in any combination. Under these conditions, the water cooler shall operate satisfactorily, maintain adequate lubrication, avoid loss of oil or other fluids, and drain without spillage. The cooler will be constantly exposed to marine environment, that is, extremes of and constantly changing ambient temperatures, high humidity in a salt laden atmosphere, and high incidence of conditions for electrolytic action of dissimilar metals. Coolers shall be highly resistant to the effects of this environment and to the shipboard conditions of vibration, roll, pitch, and shock.

3.6.2.1 Shock resistance. Each drinking water cooler and its components shall pass the high impact shock tests in accordance with MIL-S-901 for grade B, deck mounted, class II, type A equipment (see 4.3.5).

3.6.2.2 Vibration. The drinking water cooler shall meet the type I vibration requirements in accordance with MIL-STD-167-1 (see 4.3.6).

3.6.2.3 Combination of list, trim, roll, and pitch. Water coolers shall perform under list, trim, roll, and pitch conditions in accordance with DOD-STD-1399, section 301. The water cooler shall show no signs of spillage or dripping of condensate when tilted 30 degrees in any direction. The inclination and rolling shall be considered to occur in any direction.

3.6.2.3.1 Basin drain. The cooler shall free drain the cabinet top basin when the cooler is tilted 15 degrees in any direction (see 4.3.7).

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3.6.3 Capacity rating. The water cooler's capacity rating shall be based on the following conditions (see 4.3.8):

Capacity (water flow)	7-1/2 gallons per hour.
Ambient temperature	90°F.
Water in	80°F.
Water out	50°F.

3.6.4 Peak draw capacity. The outlet temperature shall be not greater than 60°F during a 15-minute period while drawing 2.75 gallons of water from the cooler. Under this condition, outlet water temperature shall be 50 ± 2°F at the beginning of the 15-minute period (see 4.3.9 and table I).

3.6.5 Maximum operating conditions. The cooler shall start and operate in an ambient temperature of 120°F with inlet water temperature of 90°F. Under these conditions at rated water flow, the water outlet temperature shall be not greater than 65°F (see 4.3.10 and table I).

3.6.6 Overload condition. The unit shall operate in an ambient temperature of 120°F, inlet water of 90°F and water drawn at the rate of 15 gallons per hour (see 4.3.11 and table I).

3.6.7 Automatic temperature control. Water temperature shall be maintained at not less than 35°F nor greater than 55°F when operating under the requirements specified in 3.6.3 (see 4.3.12).

3.6.8 Nozzle flow. With the fresh water supply at 35 to 40 lb/in^2 gauge, the stream regulator and bubbler shall deliver water at the minimum rate of 1/2 gallon per minute. With a water supply line pressure range of 20 to 70 lb/in^2 gauge, the water regulator device shall not permit the flow rate through the bubbler to deviate by more than 20 percent from the flow rate at supply line pressure of 35 to 40 lb/in^2 gauge (see 4.3.14).

3.6.9 Condensation. There shall be no visible evidence of moisture forming on the exterior of the cabinet or dripping of condensate from the piping when the water cooler is tested as specified (see 4.3.15).

3.6.10 Insulation resistance. The insulation resistance of the electrical circuit shall be 5 megohms or greater (see 4.3.16).

3.6.11 Dielectric strength. The equipment shall withstand for 1 minute a dielectric ac root mean square (rms) test voltage of 1230 volts (see 4.3.17).

3.6.12 Operating tests. The unit shall be leak free and require not greater than 7 percent more power than the nominal power input of all acceptable units when tested as specified (see 4.3.18).

3.6.13 Water system cleaning. Water system shall be flushed and cleaned to remove all foreign material, and water connections shall be covered to prevent the entrance of dirt prior to shipment of unit.

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3.6.14 Refrigerant system cleaning. Refrigerant system shall be cleaned as follows:

(a) Clean refrigerant circuit internally after leak testing. Fill with trichlorotrifluoroethane solvent in accordance with MIL-C-81302, type II. Circulate solvent through the refrigerant piping system by means of an external pump. The pump shall circulate the solvent at a velocity of not less than 20 feet per second (6.1 meters per second) through 3/8-inch (9.5-millimeter) outside diameter liquid lines.
(b) Remove internals of refrigerant control valves prior to filling the system with solvent. Open manual operating devices, if provided, during the cleaning period.
(c) Provide a circulating pump, strainers, filter, and expansion tank. Install strainer with a muslin bag ahead of pump suction. Install a refrigerant filter with replaceable filter cores or cartridges between the pump discharge and the connection to the refrigerant system. The filter cores or cartridges shall be rated at 15 micrometers absolute (remove particles larger than 15 micrometers).
(d) Stop circulation of the cleaning solvent every 2 hours, and check the muslin bag in the strainer and the filter cores or cartridges in the refrigerant filter for accumulation of contaminants and foreign particles. Replace bags, cores, or cartridges as necessary. Consider system to be clean when there is no evidence of contamination or foreign particles on the filter cores or cartridges after the 2-hour cleaning period.

3.7 Identification plates. Each water cooler shall be provided with an identification plate. The identification plate shall be located on the front of the cooler. The letters shall be Gothic capitals and numerals, and other characters shall be of similar appearance. The size of the letters, numerals, and other characters shall be as necessary for easy readability. The plates may be etched, engraved, or cast using either nickel-copper alloy brass, corrosion-resisting steel, aluminum alloy, or may be of the anodized aluminum photographic process. Plates shall contain the following information:

(a) Name of equipment.
(b) Manufacturer's name and address.
(c) Manufacturer's model.
(d) Manufacturer's serial number.
(e) Date of manufacture.
(f) Government or shipbuilder's contract or purchase order number.
(g) MIL-C-24166.
(h) NAVSEA technical manual number.
(i) National stock number.

3.7.1 Anodized aluminum photographic process. Anodized aluminum plates shall have the photosensitive silver compounds imbedded within the oxide layer. After photographic processing the image shall be sealed in the oxide layer by a chemical treatment of the layer. Stamping of additional marking information will not be permitted.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall consist of the examinations and tests as specified in table I (see 6.6.1).

4.2.1.1 First article test failure. Failure of the drinking water cooler to pass any of the tests or examinations specified in table I shall constitute rejection of the cooler. The resumption of tests and examinations shall be considered by the contracting activity after the deficiencies found have been corrected.

4.2.2 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests as specified in table I (see 6.3 and 6.6.2).

4.2.2.1 Quality conformance test failure. Production units failing to meet the quality conformance tests specified in table I shall be corrected of defective parts and components and retested until specification requirements are met prior to offering the drinking water cooler for acceptance.

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TABLE I. First article and quality conformance inspections.

Test	Requirement	Test method	Acceptance criteria	First article	Quality conformance
Refrigerant pressure and leakage	3.6.1	4.3.1	Refrigerant side shall show no evidence of leaks.	X	X
Refrigerant vacuum	3.6.1	4.3.2	Refrigerant side shall show no evidence of leaks.	X	X
Gas leakage	3.6.1	4.3.3	Water side shall show no evidence of leaks.	X	X
Hydrostatic	3.6.1	4.3.4	Water side shall be tight and no evidence of leakage or seepage shall be found.	X	X
Shock	3.6.2.1	4.3.5	(a) Dislodgement or displacement of any piece, part, or component in such a manner that would be hazardous to personnel in the vicinity of the water cooler or cause an electrical short circuit.	X	
			(b) No shearing or tearing of metallic parts that would create projections or sharp edges.	X	
			(c) No refrigerant or water leakage.	X	
Vibration	3.6.2.2	4.3.6	4.3.6	X	
Inclination	3.6.2.3	4.3.7	4.3.7	X	

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TABLE I. First article and quality conformance inspections - Continued.

Test	Requirement	Test method	Acceptance criteria	First article	Quality conformance
Rated capacity	3.6.3	4.3.8	4.3.8	X	X
Peak draw	3.6.4	4.3.9	4.3.9	X	
Maximum operating conditions	3.6.5	4.3.10	(a) The outlet water temperature shall not exceed 65 ± 2°F and the unit shall restart and continue to operate without tripping the motor overload device after each 5-minute off period.	X	
Overload	3.6.6	4.3.11	(a) The cooler shall operate without tripping the motor overload or experiencing other stoppage or breakdown during the test period.	X	
Automatic temperature control	3.6.7	4.3.12	The cooler will be acceptable if the temperature of any of the water samples is not less than 35°F or greater than 55°F.	X	X
Bubbler	3.5.7 through	4.3.13	4.3.13	X	

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TABLE I. First article and quality conformance inspections - Continued.

Test	Requirement	Test method	Acceptance criteria	First article	Quality conformance
Water flow	3.5.7.3 and 3.6.8	4.3.14	(a) The water height shall remain in the 1- to 3-inch range. (b) The water flow shall not deviate by more than plus or minus 20 percent from the initial setting.	X	
Condensation	3.6.9	4.3.15	The cooler will be acceptable if there is no visible evidence of moisture forming on the external of the cabinet or dripping of condensate from the piping.	X	
Insulation resistance	3.6.10	4.3.16	The cooler will be acceptable if the insulation resistance of the circuit is not less than 5 megohms.	X	X
Dielectric strength	3.6.11	4.3.17	The cooler will be acceptable if there is no indication of breakdown in the electrical circuit.	X	X
Operating tests	3.6.12	4.3.18	4.3.18	X	X
Electrical power	3.6.12	4.3.18.1	With equivalent inlet water and ambient temperature, the electrical power input shall be measured and compared with the input of all other units which have been tested. If	X	X

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TABLE I. First article and quality conformance inspections - Continued.

Test	Requirement	Test method	Acceptance criteria	First article	Quality conformance
			any unit requires 7 percent more power than the average of all the acceptable units, it shall be rejected.		
Water system	3.6.12	4.3.18.2	The drinking water cooler will be acceptable provided the following requirements have been met:		
			(a) The cooler shall be leak-free.		X
			(b) Each cooler shall be flushed, dried, and sealed after the operating tests.		X
Visual examination	3.5.1, 3.5.3, 3.5.3.1, 3.5.3.3	4.3.19	The following examination shall be conducted as a minimum:	X	X
			(a) Dimensions of water cooler shall be not greater than 17 inches square and 44 inches in height from top rim of the basin to the base (see 3.5.1).		
			(b) Holes shall be located in each corner of the cooler spaced 12-1/4 ± 1/8 inches center to center (see 3.5.3).		

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TABLE I. First article and quality conformance inspections - Continued.

Test	Requirement	Test method	Acceptance criteria	First article	Quality conformance
			(c) Openings shall be provided in cabinet top to permit mounting and connecting of a bubbler valve and pitcher filler (see 3.5.3.1). (d) Basin strainer openings shall not be greater in area than the area of 1/4-inch diameter holes (see 3.5.3.1). (e) Louvered openings for ventilation shall be not greater than 1/2-inch in clear width and shall not have sharp or ragged edges (see 3.5.3.3).		
Water system cleaning	3.6.13		Water system shall be flushed and cleaned to remove all foreign material.		X
Refrigerant system cleaning	3.6.14		The refrigerant system shall be cleaned of all foreign material.		X

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4.3 Test methods.

4.3.1 Refrigerant pressure and leakage test. Refrigerant containing parts of the refrigeration system shall withstand, without damage, an air and gas test of 235 lb/in^2 gauge. At this pressure, the refrigerating system shall be leak tested by means of a halide leak detector for evidence of leaks. The halide leak detector shall have a leak index sensitivity of not more than 1/2 ounce per year. Any leak detected shall be repaired and retested to insure repair has been successful.

4.3.2 Refrigerant vacuum test. At the completion of the refrigerant system pressure and leakage test the refrigerant system shall be dehydrated and evacuated to a vacuum equivalent to 5 millimeters (mm) of mercury on an absolute pressure indicator. Introduction of additives such as anhydrous methyl alcohol to reduce freezing of residual moisture will not be permitted. Upon evacuation of the unit to 5 mm of mercury, the vacuum pump shall be secured and isolated from the unit. The evacuated unit shall stand for 1 hour. An increase in pressure, corrected for ambient temperature changes, shall be cause for rejection. After a successful vacuum test, the refrigerant envelope shall be charged with the operating quantity of refrigerant and sealed.

4.3.3 Gas leakage test. The system shall be pressurized to 50 lb/in^2 gauge with 10 percent of R-12 refrigerant as a tracer gas and shall be tested for leaks with a halide leak detector with a leak index sensitivity of not more than 1/2 ounce per year. Joints shall be inspected for signs of leakage. Any leakage shall be corrected and the joint retested.

4.3.4 Hydrostatic test. The water system, including control valves, shall be tested at 150 lb/in^2 hydrostatic pressure for not less than 20 minutes. Any evidence of leakage or presence of water shall be corrected and the joint retested. The examination for leaks shall be conducted before the insulation is installed.

4.3.5 Shock test. One complete water cooler shall be subjected to high impact shock tests in accordance with MIL-S-901 for grade B, deck mounted, class II, type A equipment (see 6.3). The equipment shall be tested in an operating condition while energized and the unit running. A failure experienced during shock test shall be corrected and proven out by reshock testing.

4.3.5.1 Guidance. Guidance regarding shock tests and design of shock resistant naval equipment shall be in accordance with NAVSHIPS 0908-LP-000-3010.

4.3.6 Vibration test procedure. The vibration tests shall be running tests conducted in accordance with the test requirements for type I vibrations in MIL-STD-167-1. Unit shall operate normally after the test.

4.3.7 Combination of list, trim, roll, and pitch. The drinking water cooler shall operate for the conditions of list, trim, roll and pitch specified in 3.6.2.3. Unit shall operate normally after the test.

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4.3.8 Rated capacity test. The water cooler shall be operated for not less than 3 hours under the conditions specified in 3.6.3. The temperature control switch shall be set at 50°F. The test shall start when the initial storage tank of water has been cooled and the compressor has shut off. Water shall be drawn at the rate of 7-1/2 gallons per hour. Where a pre-cooler is installed, 60 percent of the water drawn may be used as spillage through the pre-cooler. The following shall be checked at 20-minute intervals during the test period:

(a) Ambient temperature.
(b) Inlet water temperature.
(c) Outlet water temperature.
(d) Water flow.
(e) Voltage at cooler terminals.
(f) Water cooler amperage draw.

4.3.9 Peak draw test. The peak draw test shall be conducted as specified for the rated capacity test except as follows:

(a) Operate the cooler to stabilize temperatures. Outlet water temperature shall be 50 ± 2°F.
(b) The test shall start when the condensing unit cuts out on one of its normal cycles.
(c) The test shall be conducted for 15 minutes with 2.75 gallons of water drawn at a steady rate during this period.
(d) Test readings shall be taken after 5 minutes of the start of the test and just prior to the end of the 15-minute test period (see 3.6.4).

4.3.10 Maximum operating conditions test. This test shall be conducted as specified for the rated capacity test except as follows:

(a) Operate the cooler to stabilize to the specified ambient and inlet water temperatures (see 3.6.5).
(b) Draw water at the rated flow. The water shall be discarded, but not through the drain.
(c) At the conclusion of a 1-hour period the electric power to the cooler shall be shut off for 2 minutes and again turned on for not less than 5 minutes. This shall be repeated two times. Water draw and test temperatures shall be taken during the 1-hour operating period (see 3.6.5). Repeat this procedure for each of the 2 additional 7-minute periods.

4.3.11 Overload test. This test shall be conducted as specified for the rated capacity test except as follows:

(a) The ambient temperature shall be 120°F.
(b) The inlet water temperature shall be 90°F.
(c) The water shall be drawn and discarded, not through the drain, at a constant rate of 15 gallons per hour.
(d) Water flow and temperature readings shall be taken.
(e) The test shall be conducted until steady conditions are observed for not less than 3 hours.

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4.3.12 Automatic temperature control test. The temperature control shall be set at its lowest setting and the cooler operated in an ambient temperature of 90°F and an inlet water temperature of 80°F. The cooler shall be operated until the temperature control stops the compressor. Immediately after the compressor stops, water shall be drawn for 30 ± 5 seconds. The water outlet temperature shall be taken during this period. Water shall also be drawn immediately after the compressor starts and the water temperature taken. This procedure shall be repeated three times. The temperature control shall be reset to maintain a water temperature of 50°F at the conclusion of this test (see 3.5.9).

4.3.13 Bubbler examination. The bubbler shall meet all the requirements of 3.5.7, 3.5.7.1, and 3.5.7.2.

4.3.14 Water flow test. The water flow test shall be performed as follows (see 3.5.7.3 and 3.6.8):

(a) Adjust the inlet water pressure to 40 lb/in^2 and flow rate to not less than 1/2 gallon per minute.
(b) Measure the water height above the guard and flow rate.
(c) Repeat the test at 20 and 70 lb/in^2 inlet water pressure.

4.3.15 Condensation test. The cooler shall be operated as specified for the rated capacity test (see 4.3.8) and with an ambient relative humidity of not less than 50 percent. Temperatures, flow, and relative humidity readings shall be taken during this test.

4.3.16 Electrical insulation resistance test. With the water cooler at ambient temperature, the electrical insulation resistance shall be as specified in 3.6.10. The measuring of insulation shall be made by connecting the line terminals together, manually closing switches and controls, and measuring the resistance between terminals and ground. The electrical insulation resistance shall be measured with a 500-volt insulation resistance meter. The time of test voltage application shall be not less than 60 seconds.

4.3.17 Dielectric strength test. The dielectric strength test shall be conducted with the motor at ambient temperature, the line terminals together, and manually closing switches and controls. A dielectric ac rms test voltage of 1230 volts shall be applied for 1 minute between line terminals and ground (see 3.6.11).

4.3.18 Operating tests. The units shall be connected to the water supply and operated for not less than 1 hour with controls set for specified operating conditions. Following the 1-hour operating test, the controls shall be tested for proper setting and satisfactory operation.

4.3.18.1 Electrical power. See table I for acceptance criteria.

4.3.18.2 Water system. See table I for acceptance criteria.

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4.3.19 Examination. Each unit shall be examined for conformance to the following requirements:

(a) The compressor valve seal caps, shall be tight for shipment and all other seal caps in the refrigerant system provided with a gasket shall be tight against refrigerant leakage.
(b) Bolts, nuts, and screws shall be tight; equipment and parts shall be properly fastened and secured.
(c) No parts shall be fractured, split, torn, dented, or otherwise damaged so as to affect serviceability.
(d) Cold lines shall be properly insulated (see 3.5.6).
(e) The limiting and mounting dimensions shall be as specified (see 3.5.1 and 3.5.3).
(f) Electrical wiring shall be properly installed and connected (see 3.6.10 and 3.6.11).
(g) There shall be no sharp or ragged edges on the sheeting that may be injurious to personnel.
(h) Drinking water cooler shall be drained.
(i) Fittings for the external connections shall be properly secured to the frame (see 3.5.6).
(j) Refrigerant system shall be free of refrigerant leakage.
(k) Controls or switches shall be properly set and in operating order.

4.4 Inspection of packaging. Packages and packs, and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Packaging requirements. Water coolers shall be preserved level A or C, packed level A or B as specified (see 6.2) and marked as specified in 5.4 and shall include bar codes and applicable packaging acquisition options therein as specified (see 6.2). In addition, for Navy acquisitions, the following applies:

(a) Navy shipboard stowage fire-retardant requirements.

(1) Lumber and plywood. When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container construction members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.

Level C - Type I - non-weather resistant.
Category 1 - general use.

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(2) Fiberboard. When specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packaging forms shall meet the flamespread and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Preservation.

5.2.1 Level A. Each complete unit (see 3.5.1) shall be unit protected and packaged in accordance with method I of MIL-P-116 and as follows: The water system shall be thoroughly drained and blown out by the application of clean, dry compressed air. Openings including the bubbler and basin drain shall be sealed to prevent entrance of dirt and moisture. Drive belts shall be removed, exposed uncoated ferrous metal surfaces of pulleys and shafts cleaned and coated with primer conforming to TT-P-664. When the primer is thoroughly dry, drive belts shall be remounted in place with tension on the belts relaxed. Compressor and motor mounting and shipping bolts, nuts, and screws shall be protected from corrosion by the application of preservation compound conforming to grade 4 of MIL-C-16173.

5.2.2 Level C. Preservation of each unit shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be levels A, B, C, or commercial as specified (see 6.2).

5.3.1 General requirements for levels A, B, C. Containers selected (see 5.3.2) shall be of minimum weight and cube consistant with the protection required, of uniform size, and contain identical quantities.

5.3.2 Levels A, B, C. Water coolers preserved as specified (see 5.2) shall be packed in exterior shipping containers in accordance with MIL-STD-2073-1, table VII of appendix C, for the level of packing specified (see 5.3). Unless otherwise specified (see 6.2), container selection including container options shall be the contractor's option.

5.3.2.1 Closure, gross weight, and waterproofing.

5.3.2.1.1 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and domestic nonweather-resistant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.2 Weight. Wood, plywood, and wood cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1, appendix F, or the applicable container specification or appendix thereto.

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5.3.2.1.3 Waterproofing. Unless otherwise specified (see 6.2), level A and, when specified (see 6.2), level B shipping containers shall be provided with caseliners, linings, wraps, or shrouds in accordance with the waterproofing requirements of MIL-STD-1186.

5.3.3 Commercial. Water coolers preserved as specified (see 5.2) shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage, and storage.

5.4 Marking.

5.4.1 Levels A, B, C, and commercial. In addition to any special marking required (see 6.2), interior (unit) packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-2073-1, appendix F, and shall include bar codes and applicable packaging acquisition options herein as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The drinking water coolers provide chilled drinking water in various locations on Naval ships.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) When first article is required (see 3.1).
(d) When a refrigerant other than R-12 is specified (see 3.5.4).
(e) Whether pitcher fillers for drinking water coolers are required (see 3.5.7.2).
(f) Level of preservation and packing required (see 5.1).
(g) If fire-retardant materials are not required (see 5.1(a)).
(h) Packing requirements (see 5.3).
(i) If waterproofing materials are not required (see 5.3.2.1.3).
(j) When container selection is other than the contractor's option (see 5.3.2).
(k) Marking and special marking requirements (see 5.4).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2 and 3.2.1	DI-E-2121	Certificate of compliance	-----
3.3	DI-R-7079	Reliability program plan	-----
3.5.1 and appendix	DI-DRPR-80651	Engineering drawings	-----
4.2.2	UDI-T-23732	Procedures, test	-----
4.2.2	DI-T-2072	Reports, test	-----
4.3.5	DI-ENVR-80708	Shock test report	-----

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the

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Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Subcontracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.7.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.8 Shock. The general information given in MIL-S-901 should be carefully considered in the design of equipment. This information is also used by the drawing review activity as a factor in approval of drawings required for production equipment.

6.9 Materials. Where Unified Numbering System (UNS) in accordance with SAE J 1086 are listed in section 3 in conjunction with listed Military, Federal and commercial specifications, reference to a specific UNS designation will permit the use of any one of the many alloy specifications and standards listed in SAE J 1086 for that UNS number.

6.10 Subject term (key word) listing.

Basin strainer
Condensing unit
Hermetic compressor
Liquid receiver

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4110-N424)

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APPENDIX

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that shall be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWING CONTENTS

30.1 Drawings. Drawings shall contain the information specified in 30.1.1 through 30.1.5. Separate drawings need not be furnished for each category. Parts lists, bill of material lists, and so forth, shall be included on equipment drawings. Certification data drawings should include diagrams and drawing lists.

30.1.1 Assembly drawings. Assembly drawings shall be provided for the complete drinking water cooler. The drawings shall illustrate the design, construction, operation (or function), identity of parts, and performance data. Section or partial views either on the assembly or detail drawings shall show attachment and fastenings of cabinet hardware, supports, evaporator mountings and support and stiffening of cabinet or panels where provided. Subassembly drawings shall be furnished where assembly drawings do not adequately describe and identify subassembly parts and components. The drawings shall reflect compliance with equipment specification requirements. Drawings of the following components shall be provided:

(a) Cabinet.
(b) Compressor and motor.
(c) Fan motor.
(d) Condenser.
(e) Condenser fan assembly.
(f) Receiver.
(g) Evaporator assembly.
(h) Controls, valves and electrical components.

30.1.1.1 Assembly drawings shall show the following information:

(a) Outline dimensions.
(b) Mounting, attachment and connection dimensions.
(c) Methods and sizes of fastenings or connections.
(d) Weight and center of gravity of each complete unit.
(e) Clearances required for operation and maintenance.
(f) Data necessary to permit installation of the equipment without the contractor's assistance.

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30.1.2 Sectional assemblies. Sectional assemblies shall be provided as required to supplement the data on assembly drawings. Running clearances shall be shown. The list of material shall include every part required in the assembly, including those parts not required to be detailed. The list of material shall include type of material, material specification, manufacturer and manufacturer's part number. This may necessitate some side or partial views in order to show parts not otherwise shown in the main section view.

30.1.3 Detail drawings. Detail drawings shall be furnished for all parts and subassemblies necessary for evaluation of the equipment and all parts required for maintenance and overhaul of the equipment. Subassemblies whose parts cannot be acquired or serviced individually should be shown as a single part. Multi-detail drawings are preferred but mono-detail drawings may be furnished. Drawings are not required for those parts which are in common commercial use and can be referenced to commercial standards.

30.1.4 Certification data sheets. Certification data sheets (drawings), shall be provided. The drawings shall include a reference drawing list of all the assembly, subassembly and detail drawings of the equipment and components being furnished on the contract or order. Diagrams shall also be provided. The diagrams shall show by symbolic representation all fluid piping, pipe sizes and connection identification, electrical interconnections, components, accessories and controls plus connections required by others for operation. For example, the refrigerant diagram will include the associated condensing unit, controls, valves, piping accessories and evaporator as applicable. The electrical wiring diagram will include and show circuits between motors, controls, switches and accessories as applicable for understanding the operation sequence. Symbols used for equipment in diagram drawings shall be given a piece number and identified in the list of material with the following information:

(a) Piece number.
(b) Quantity required.
(c) Descriptive name.
(d) Manufacturer's name.
(e) Manufacturer's identifying model or part number.
(f) Weight.
(g) Characteristics.

30.1.5 Characteristic data. The characteristics shall include ordering information necessary to limit or supplement data on referenced drawings. Complete characteristics for the drinking water cooler as well as for all major parts or components including the compressor, condenser, receiver, cooling coils, fan and fan motor, switches, accessories and expansion device as applicable shall be furnished.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-24166B(SH)	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
COOLER, DRINKING WATER, MECHANICALLY REFRIGERATED, SELF-CONTAINED, NAVAL SHIPBOARD

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC): Mr. John Merold (SEA 56Y15) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: 703-602-0741
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	(2) AUTOVON IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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